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IrradianceNet: Spatiotemporal deep learning model for satellite-derived solar irradiance short-term forecasting

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ABSTRACT

The presence of clouds is widely identified as the primary uncertainty in current surface solar global horizontal irradiance (GHI) forecasts. Despite a wealth of historical satellite-derived irradiance observations, only limited research has investigated this problem from a purely data-driven perspective, something that has seen tremendous success in related domains such as radar- and satellite-based precipitation short-term forecasting. This paper presents IrradianceNet, a novel satellite-based neural network for spatiotemporal forecasting of surface solar irradiance up to 4 h in the future over Europe. Our method is fully data-driven and needs no post-processing or calibration based on sparse ground-based measurements of irradiance. We demonstrate superior forecasting performance compared to several persistence models, the *TV-L*[1] algorithm, and ERA5 reanalysis data for satellite-derived solar irradiance using the European SARAH-2.1 dataset. We also validate these results using ground-based pyranometer observations from the Baseline Surface Radiation Network. Our conclusions remain unchanged when we account for hourly and monthly seasonality. Finally, applying a simple cloud mask scheme, we demonstrate that our performance improvement arises due to a considerable reduction in cloudy pixel errors. This is initial evidence that purely data-driven methods might better approximate and infer future cloud dynamics and their impact on surface solar irradiance.

1. Introduction

To meet the growing demand for renewable energy, solar energy poses a promising solution. The increasing integration of solar energy has led to unprecedented scientific research into surface solar (global) irradiance forecasting, which is the main input for photovoltaics (PV) production (Antonanzas et al., 2016). Despite the well-established economic and environmental benefits of PV, there are still considerable risks to the stability of the power grid. The primary risk arises from solar being a fluctuating energy resource, which necessitates accurate solar irradiance forecasts. This resource variability is explained by a deterministic component and a stochastic component. The deterministic component represents the daily and seasonal variations arising due to the movement of the Earth with respect to the Sun. The stochastic component is explained by several stochastic events that impact surface solar irradiance, such as the frequency of clouds relative to their optical depth and height and the composition of the turbidity of the atmosphere. While physical laws can describe cloud movement and formation, these processes are inherently stochastic due to turbulent processes making them extremely difficult to model (Larson, 2013). For these reasons, clouds represent the primary uncertainty in current solar irradiance forecasts. As a result, most solar irradiance forecasting algorithms focus specifically on improved modeling of cloud dynamics, identified as one of the major challenges in widespread PV integration across the globe (Antonanzas et al., 2016).

Solar irradiance forecasting can be traced all the way back to the late 19th century when numerical weather prediction models (NWP) were first developed (Yang et al., 2018). Since then, numerous methods have been developed to improve forecast horizons spanning from very short-term forecasts to multiple days ahead (Lorenz et al., 2009). Generally speaking, forecasts based on satellite data are considered best for short-term forecasting (0-5 h ahead) (Perez et al., 2010). Beyond this forecast horizon, statistical methods that combine NWP model output with other data types such as satellite tend to perform the best (Perez et al., 2010).

In recent years, spatiotemporal deep learning models have gained tremendous interest in the atmospheric forecasting community. This started in 2015 where it was shown that modern deep learning methods such as convolutional long short-term memory (ConvLSTM) networks

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could outperform the state-of-the-art optical flow method for precipitation forecasting implemented at the Hong Kong Observatory (Xingjian et al., 2015). However, outside precipitation forecasting, these data-driven methods have not seen the same level of adaptation in the atmospheric community, such as the domain of deterministic short-term solar irradiance forecasting.

In the deterministic short-term solar irradiance forecasting literature, most data-driven methods to date that utilize satellite-derived irradiance fields typically incorporate ground measurements in some way to calibrate their model (Pérez et al., 2021). As stated by Pérez et al. (2021), this can be problematic in regions where ground measurements are either unavailable or unreliable. Several recent works have tried overcoming this limitation. For example, Larson and Coimbra (2018) use support vector regression to calibrate their satellite-based forecast based on only a few ground-based measurements, which reduces, but does not eliminate, the need for ground observations. In another work by Lago et al. (2018), they develop and fit a deep neural network based on clear-sky irradiance, satellite-derived GHI, and NWP data. However, similar to Larson and Coimbra (2018), their model still needs ground-based data for calibration.

In another work by Urbich et al. (2019), the authors demonstrate using cloud motion vectors (CMV) to extrapolate future satellite irradiance frames, a popular method in the solar irradiance short-term forecasting literature and analogous to advection equations in meteorology (Urbich et al., 2019). The primary drawback of CMV is the inability to model cloud formation and dissipation, thus requiring extensive post-processing to improve these processes (Ayet and Tandeo, 2018). Nevertheless, Urbich et al. (2019) demonstrate better results compared to several NWP-based models and the simple persistence model and provide a benchmark for the SARAH-2.1 dataset. As a result, we replicate the experimental setup for the *TV-L*[1] used by Urbich et al. (2019) to serve as an additional benchmark model to IrradianceNet.

Finally, the most similar work to ours is by Pérez et al. (2021). Their method is an extension of the model developed by Lago et al. (2018), in which they eliminate the need for ground-based measurements completely. This is achieved by only training on past satellite-derived GHI observations using a convolutional neural network architecture with secondary input data included at a later stage as input to fully connected layers. Our work differs from Pérez et al. (2021) in several important ways. First and foremost, they train their model directly on total satellite-derived GHI in contrast to our method, where we train our model exclusively on the effective cloud albedo and later incorporate clear-sky irradiance. We do this to focus exclusively on cloud dynamics, the primary uncertainty in solar irradiance forecasting models as stated earlier. Second, we focus on the entire geographical area of Europe for our deep learning model instead of only specific areas with neighboring pixels. As a result, we can a) predict irradiance anywhere in Europe, thus extending the potential users of our forecast to include system operators or even residential rooftop PV end-users, and b) obtain significantly more observations. The latter should, in theory, imply better generalization since our model has been trained on multiple areas with different climates. In addition, by using the effective cloud albedo, we obtain data that is scale-invariant to the angle of the Sun, which is not the case when using GHI directly. Convolutional neural networks are not scale-invariant by default (Goodfellow et al., 2016), and as a result, this can make the cloud motion estimation negatively biased. We discuss this point further in Section 2. Finally, we also incorporate recurrence as a central feature in our model, which is critical to success in the related field of video prediction (Villegas et al., 2019) and also noted as an avenue for future work by Pérez et al. (2021).

More specifically, we propose a novel patch-based spatiotemporal deep learning approach for satellite-derived solar irradiance forecasting called IrradianceNet. Besides geostationary satellite images, we incorporate descriptive features such as topographical information, coordinates, and calendar features. As a result, our model does not rely on NWP output or geographically scattered ground-based observations. This makes our method inherently global. To ensure the reproducibility of our method and results, we have supplied the code and several examples of our model in action at https://github.com/holmdk/IrradianceNet.

We demonstrate that IrradianceNet can outperform the fifth generation European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) data, and state-of-the-art optical flow and persistence models for lead times up to 4 h in the future. This result is also confirmed using the deterministic evaluation framework proposed by Yang et al. (2020) for four independent ground-based measurements of surface solar irradiance from the Baseline Surface Radiation Network (BSRN) (Ohmura et al., 1998).

We also show that our model can produce an accurate cloud mask forecast with some simple thresholding rules, which is another way of validating an irradiance forecasting scheme (Urbich et al., 2019). Additionally, we investigate making IrradianceNet probabilistic using some minor modifications (Sønderby et al., 2020), providing valuable information to energy market participants via uncertainty quantification (Pinson et al., 2007; Iversen et al., 2014). To our knowledge, this is one of the first attempts at developing a general deep learning model that can provide both deterministic and probabilistic forecasts with only minor modifications. The details of this will be discussed further in Section 3. As a final experiment, we investigate changing several parameters of IrradianceNet in an ablation study.

In Section 2, we present the data sources used in this study. Next, we describe our proposed method called IrradianceNet in Section 3. Then, in Section 4, we propose our experimental and evaluation design. In Section 5, we discuss the results from our experiments and finally conclude in Section 6.

2. Data

We start by defining the input variables used by our model in Section 2.1 and 2.2, followed by the external BSRN observations in Section 2.3, and finally, the NWP data used for benchmarking in Section 2.4. We also discuss the work required to use our proposed method in an operational short-term forecasting application in Section 2.1.1.

2.1. Effective Cloud Albedo

In order to generate short-term forecasts of solar irradiance, it is well established that satellite is one of the most optimal data sources (Antonanzas et al., 2016). Satellite also allows us to obtain a wealth of historical observations, a requirement for data-driven methods. Given that clouds have the largest impact on short-term solar irradiance, we are primarily interested in improving short-term forecasts of cloud properties (Urbich et al., 2018). The effective cloud albedo (CAL) or cloud index seems a reasonable choice for this purpose, as it a) directly measures the impact of clouds on the surface solar irradiance and b) filter the ground albedo from the solar irradiance measurement, thus enabling our model to focus exclusively on cloud dynamics (Urbich et al., 2018). This is advantageous compared to using satellite-derived irradiance fields directly, as stated in Section 1, as the model also needs to learn to forecast clear-sky irradiance like in Pérez et al. (2021).

The effective cloud albedo is derived using the reflectivity from the visible bands of satellites in a similar fashion to the popular Heliosat method, which implies no external model information is needed (Mueller et al., 2011). Importantly, using this method, we can establish an almost linear relation between solar irradiance and the effective cloud albedo given by the Heliosat relation, which is based on the law of energy conservation (Mueller et al., 2011). The reason it is not strictly linear relates to empirical corrections being placed upon certain CAL values, such as CAL > 0.8, where the saturation and absorption effects in optically thick clouds need to be taken into account. These corrections are described in detail by Pfeifroth et al. (2019). The relationship between solar irradiance and the effective cloud albedo is thus defined as

$$SIS = (1 - CAL) \cdot SIS_{clear}, \quad (1)$$

where SIS is the total surface solar irradiance and SIS_{cs} is the clear-sky solar irradiance. For notational purposes, we will redefine the first part of the right-hand side as

$$k = (1 - CAL). \quad (2)$$

This effectively means that for any pixel in the satellite image, we can convert our forecast of CAL directly to an estimate of surface solar irradiance measured in watts per square meter using Eq. 1. This approach aligns with the conventional time series procedure for solar irradiance forecasting, where one eliminates the deterministic components to make the series stationary and then model the time series of the stochastic term (Linguet et al., 2016). After modeling the stochastic term, the deterministic part is then reintroduced to obtain the combined time series of solar irradiance. If we relate Eqs. 1 and 2 to the deterministic and stochastic components of solar irradiance, it can be seen that SIS_{clear} refers to the deterministic component and k to the stochastic component (not accounting for turbidity composition).

In this paper, we use the CAL derived from the European Meteosat Second Generation (MSG) satellites operated by the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT). This dataset is called SARAH-2.1 (Urbich et al., 2019) and is available from 1983–01-01 to 2017–12-31 in 30-min increments with a spatial resolution of 0.05. In Urbich et al. (2018), they use a clear-sky algorithm called SPECMAGIC NOW, which spectrally resolves irradiance from the visible channel of satellite images. Unfortunately, we do not have access to the SPECMAGIC NOW algorithm, but using the parameterizations from Pfeifroth et al. (2019) and simply re-arranging Eq. 1 to solve for SIS_{clear}, we can deduce the spectrally resolved clear-sky irradiance to generate a dataset of satellite-derived clear-sky irradiances. We also experimented using a simple clear-sky model like the Ineichen clear-sky model (Ineichen and Perez, 2002), which yielded similar results.

We conduct preprocessing to account for missing observations similar to Pfenninger and Staffell (2016). This includes interpolating short-term missing observations from neighboring values (below 6 h) and for longer periods using the same dates from preceding years to fill in any missing values.

2.1.1. Operational Feasibility

While the SARAH-2.1 dataset is essentially a post-processed product available in historical data records, it is possible to derive the effective cloud albedo in an operational setting with only limited dissemination lag. There are two ways of achieving this, which we will briefly outline. The first method is the simplest and only requires using the OpenData weather service provided by the German Weather Institute (Deutscher Wetterdienst (DWD), 2021), where one can download SIS and SIS_{clear} around 2-3 min after DWD has collected the satellite images [1]. Then, having these two quantities, one can simply derive the effective cloud albedo by rearranging Eq. 1

$$CAL = 1 - \frac{SIS}{SIS_{clear}}. \quad (3)$$

The other approach is a bit more involved but still relatively simple. As CAL is derived using a modified version of the Heliosat method, we already established in Section 2.1 that no external mode information is needed. Thus, the only quantities needed in addition to pixel counts are the maximum reflectivity and minimum reflectivity within a given month (Pfeifroth et al., 2019). As we do not know these quantities in an operational setting, a simple modification could be using the previous 30-day running maximum and minimum reflectivity instead. Due to these reasons, we believe there is ample potential for using this dataset and our model in an operational setting, which we discuss further in Section 3.6.

[1] Richard Müller and Jörg Trentmann (personal communication, August 13 and August 16, 2021)

2.2. Descriptive data

As mentioned in Section 1, we supply our satellite observations with several meta-features. The static features include a 2 D elevation map (Mitchell, 2014), and latitude and longitude coordinates for each pixel in the satellite images. This should improve modeling effects such as orographic lifting, as the model should, in theory, be able to encode and decode spatiotemporal patterns conditioned on the specific area under investigation. We also include deterministic features to help capture seasonality effects: the month, day, and hour for each observation.

2.3. Ground stations

We include ground-based observations of solar irradiance measurements to serve as an additional validation of our results. As these are completely independent of the SARAH-2.1 dataset, they could, in theory, reveal inaccuracies in the satellite-derived solar irradiance measurements. As a result, including this validation set essentially eliminates any potential bias from the SARAH-2.1 dataset that might propagate into our IrradianceNet model and subsequent performance evaluation. Nevertheless, since the SARAH-2.1 dataset has already been evaluated against ground-based stations with good results (Urbich et al., 2019), we do not expect this to impact our results dramatically. The chosen ground-based stations consist of four stations from the well-known Baseline Surface Radiation Network (BSRN). In advance, we note that a small and positive bias is actually found in Yang (2018) specifically for these four sites in the SARAH-2.1 dataset described in Section 2.1, which we expect to propagate into our data-driven model. The four stations are all located in Europe and match the same set used by Urbich et al. (2019). These are all based on pyranometer measurements of global irradiance and therefore considered highly accurate. The four stations are located in different regions in Europe, covering both the cloudy and unstable climates of northwest Europe and the more stable and mountainous regions of southern Europe.

2.4. NWP Output

We include ERA5 reanalysis data as a benchmark model. In Perez et al. (2013), the authors conducted a comparison of several global, multiscale and mesoscale NWP models in both North America and Europe for solar irradiance forecasting and found that IFS irradiance forecasts showed a higher accuracy compared with WRF models and other global mesoscale models such as the Global Forecast System (GFS) model.

The primary disadvantage of NWP-based short-term forecasting is the coarse spatial resolution, few model updates per day, and considerable dissemination lag for operational models. This typically implies satellite-based extrapolation methods with the finer spatial resolution and almost real-time update rate generally outperform global NWP methods (Perez et al., 2010), as stated in Section 1. Using ERA5 reanalysis data, however, we do eliminate the impact of dissemination lag.

3. IrradianceNet

This section will describe the methodology behind IrradianceNet, our novel neural network-based spatiotemporal predictive model.

3.1. Input

The input to IrradianceNet has already been described in Section 2.1 and 2.2. It corresponds to a four-dimensional tensor of size $[c, t, h, w]$ where c is the number of input channels, t is time, h is height, and w is

width. The area under investigation is Europe, which yields us images of size 512×512 pixels. The number of input channels, c, is equal to 7, which at each timestep t consists of the effective cloud albedo image, an elevation map, longitude, latitude, hour, day, and month. As the spatial dimension varies between these features, we either replicate the feature values along the missing dimension, similar to Sønderby et al. (2020), or select a nearest-neighbor lookup to have the same $h \times w$ size as the effective cloud albedo images. An overview of our input data can be seen in Fig. 1. The final resolution of our dataset is $[7, t, 512, 512]$, where t is a hyperparameter we set prior to training.

3.2. Patch-based cropping

Due to the relatively large size of our input images, we employ a patch-based approach to the spatiotemporal modeling procedure as seen in other applications with large input images (Sønderby et al., 2020). This involves dividing each of our 512×512 images into 128×128 patches, which leaves us with 4×4 of these 128×128 patches. This has the added benefit of a) reducing memory footprint of the memory-intensive ConvLSTM layers by having them train on each patch independently, and b) we can select random patches of 128×128 during training. The latter is similar to the random cropping technique typically employed for achieving invariance with data augmentation techniques, thus essentially increasing the size of our dataset by a factor of $4 \times 4 = 16$. During inference, we predict all patches independently, concatenate them and reshape them to their original 512×512 resolution. Patch-based cropping typically results in considerable border artifacts between patches, which in our case was reduced considerably by using linear interpolation techniques along the patch borders. In Section 5 we will visualize and discuss these border effects further.

3.3. Probabilistic output encoding

One simple approach for generating probabilistic forecasts from a neural network is to discretize a continuous output into a categorical distribution. To achieve this, we map the effective cloud albedo values ranging from $[0; 1.2]$ into 240 discrete bins, where each bin corresponds to a 0.005 interval. Having probabilistic predictions, we can then select the class with the highest probability from our model output to obtain an effective cloud albedo forecast, or we can estimate the probability of CAL values above or below a certain threshold simply by summation. This also enables us to quantify uncertainty in our model predictions, which is infeasible for the deterministic version of IrradianceNet.

3.4. Network Design

We introduce a novel spatiotemporal autoencoder architecture with three ConvLSTM layers in both the encoder and the decoder network. For every layer in our encoder network, we apply a ConvLSTM cell to iterate over all timesteps and reduce the height and width dimension of the patches by a factor of 2 while increasing c in 32 increments, leaving t unchanged. This is achieved by passing the final hidden state of each layer as the input to the next. This procedure is repeated for three ConvLSTM cells, which leaves us with a tensor of size $[128, t, \frac{h}{4}, \frac{w}{4}]$. Next, we apply three ConvLSTM layers to perform the upsampling operation found in the decoder part of the network, essentially performing spatiotemporal forecasting. This returns a tensor of size $[c, t, h, w]$ where $c = 1$. Next, we apply a 3 D CNN in the final layer of our network to perform either a) classification by mapping the $c = 1$ dimension into the 240 categorical distribution, i.e., $c = 240$, or b) regression simply by keeping $c = 1$ fixed. This has the benefit of being able to condition on all output time steps rather than individually using 2 D CNNs at each time step. An overview of our entire network architecture can be found in Fig. 1.

3.5. Training details

For the deterministic IrradianceNet, we use mean-squared error as our loss function. We also experiment using the $\ell 1$ norm either in isolation or combined with the mean-squared error, but this did not improve our results. For the probabilistic IrradianceNet, where our task is categorical classification, we use cross-entropy as our loss function to optimize the predictive distribution of the 240 output bins. We use leaky ReLU as the activation function in all layers. Similar to most recurrent neural networks, we initialize all hidden states with zeros. For the CNN cells, we initialize them with the popular Xavier initialization (Glorot and Bengio, 2010). We use the Adam optimizer with an initial learning rate of 0.002 and gradually decrease it by half once the validation loss stops improving beyond a small threshold for at least five consecutive epochs, also called reduce learning rate on plateau. We train a model for each lead time independently, which generally yields more stable training and better results than an aggregate model. We use four previous time steps as input to IrradianceNet, but we also experiment using only two-time steps in our ablation study in Section 4.3. Training a model for each lead time takes about four days per lead time using two NVIDIA RTX 2080-TI cards. For more details on the training, validation, and test setup, we refer to Section 4.

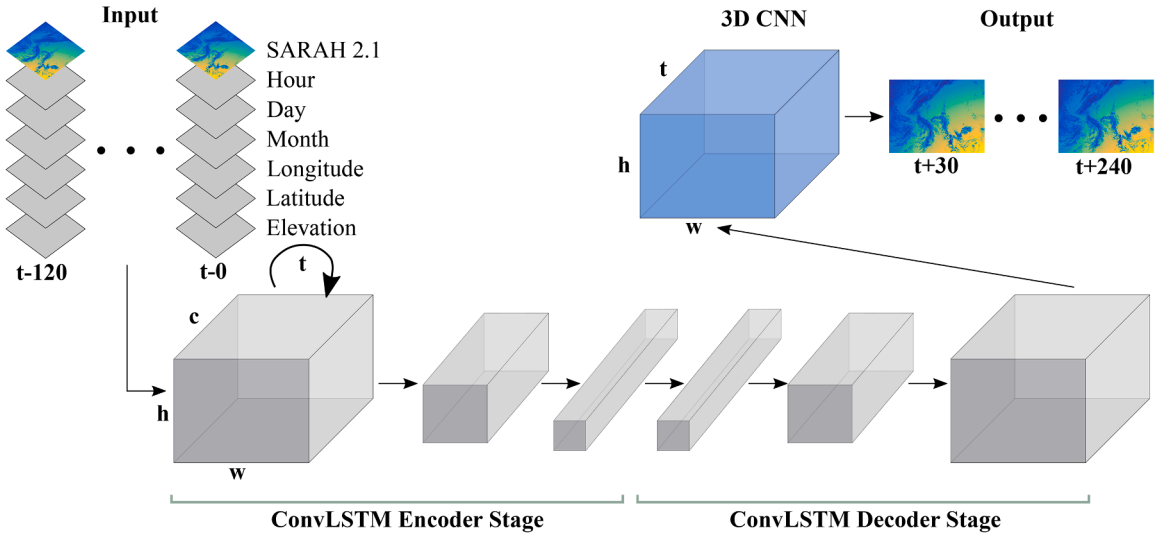


Fig. 1. Overview of the IrradianceNet architecture, where t is lead time measured in minutes, w is width, h is height and c is depth.

3.6. Operational setting

In an operational setting, we already established that the input data, CAL, can be derived in a few minutes after collecting the raw satellite images. For our full IrradianceNet model, the processing time for one satellite image is around 0.56 min on one Nvidia RTX 2080-TI card as seen in Table 5. Compared with the deep learning model from Pérez et al. (2021), where the delay is a minimum lead-time of 30 min, we can generate a solar irradiance forecast with a lead-time of only around 5 min.

4. Experiments

This section describes the experimental setup used, including a discussion of the chosen benchmark models in Section 4.1, the performance evaluation setup in Section 4.2, and an ablation study in 4.3.

First, we evaluate all models using the SARAH-2.1 dataset as ground truth. We have divided the dataset into a train, validation, and test set. The training set consists of the years 2010–2014, the validation set includes 2015, and the test set includes 2016–2017. All datasets have a temporal resolution of 30 min and a spatial resolution of 512×512. The task is multi-step-ahead prediction for up to four hours in the future. Unless otherwise noted, all results presented in Section 5 will be based on the held-out test set. In addition to this, we include ground-based solar irradiance measurements from the widely recognized BSRN as described in Section 2.3.

4.1. Benchmark models

We evaluate IrradianceNet relative to several baseline models for irradiance short-term forecasting. These include a) the *TV-L*^{1} Optical Flow algorithm for future frame extrapolation (Urbich et al., 2018; Urbich et al., 2019), b) ERA5 reanalysis, c) naive persistence, and d) smart persistence. For our BSRN evaluation, we also include the prominent climatology-persistence model recommended by Yang et al. (2020). The configuration of *TV-L*^{1} and the persistence methods will be discussed shortly. The details of ERA5 and IrradianceNet were already outlined in Sections 2.4 and 3, respectively.

4.1.1. Persistence Models

One of the most common and simplest benchmark models in the solar irradiance forecasting literature is called a persistence model. The simplest variant is the "naive persistence", which assumes the forecasted solar irradiance is equal to the latest observation of solar irradiance. As stated by Antonanzas et al. (2016), this model is only recommended for time series that are stationary. A better alternative to the naive model is called the "smart persistence" model. It works for both intra-hour applications and longer time horizons and is considered among the top models for solar irradiance forecasting due to the complexity of modeling cloud processes (Kumler et al., 2018). It works by decomposing the solar irradiance forecast into a stationary term and a stochastic term. The stationary term is typically the clear-sky irradiance, and the stochastic term is the cloud-induced solar irradiance. At time t, we can denote the full solar irradiance, I_t, as

$$I_t = I_t^{cs} + I_t^{ci} \tag{4}$$

where I_t^{cs} is the clear-sky solar irradiance and I_t^{ci} is the cloud-induced solar irradiance. To generate a forecast of the next step solar irradiance, we choose the following approach (Kaur et al., 2016; Sengupta et al., 2017)

$$I_{t+1} = I_{t+1}^{cs} \cdot \frac{I_t}{I_t^{cs}} \tag{5}$$

where $\frac{I_t}{I_t^{cs}}$ is typically called the clear-sky index.

If we relate the two terms of this equation back to the definition of SIS and k in Eq. 1 and 2, respectively, we can rewrite our equation as

$$I_{t+1} = I_{t+1}^{cs} \cdot k_t, \tag{6}$$

where I_{t+1}^{cs} is equal to SIS_{clear} and k_t is equal to k. This is the final definition of the smart persistence model, where it can be observed that the cloud-induced changes from k_t are advected into the future, and the clear-sky irradiance I_{t+1}^{cs} is estimated via a clear-sky model similar to the one described in Section 2.1.

Finally, it is noted by Yang et al. (2020) that using the optimal convex combination of climatology and persistence, called climatology-persistence, is highly recommended in deterministic forecasting studies due to its potentially superior performance compared to the smart persistence model. We will use the climatology-persistence model for our BSRN evaluation, and we refer to Yang et al. (2020) for more details regarding this model.

4.1.2. TV-L^1 Optical Flow

As stated in Section 1, we will implement the *TV-L*^{1} algorithm used by Urbich et al. (2019), considered state-of-the-art for the SARAH-2.1 dataset. In their paper, the authors optimized 11 different hyperparameters based on a grid search with a modified area-under-the-curve calculation based on the absolute bias as a function of forecast time for 21 different configurations. We widened the parameter search to include 360 different combinations relative to the optimal hyperparameters found by Urbich et al. (2018, 2019) and found a slightly different set of optimal hyperparameters for the validation set. We kindly refer to https://github.com/holmdk/IrradianceNet for further details.

4.2. Performance Metrics

A key gap in developing solar forecasting models is the unavailability of a consistent and robust set of metrics to measure and assess forecasting accuracy. This is because different researchers use improvements described by different metrics as their evaluation criteria (Zhang et al., 2015).

For our spatiotemporal evaluation, we consult several extensive evaluation studies (Lorenz et al., 2009; Perez et al., 2010; Yang et al., 2020). For our gridded evaluation using the SARAH-2.1 dataset as target ground-truth, we use the root mean squared error (RMSE) and Mean Absolute Error (MAE). We note that RMSE is considered the best metric for evaluating gridded irradiance forecasts (Yang and Bright, 2020), and as such, we will emphasize this metric when discussing our results in Section 5. To enable comparisons across spatial and temporal scales irrespective of potential outliers, we normalize RMSE relative to the interquartile range at the 5% and 95% percentile over actual irradiance observations, denoted $Q_{.05}$ and $Q_{.95}$. We denote this metric as the relative RMSE (rRMSE) given by

$$rRMSE = \frac{RMSE}{Q_{.95} - Q_{.5}}. \tag{7}$$

There are also limitations for RMSE and MAE despite their popularity. For example, these metrics are only unbiased if the forecast errors follow a Gaussian distribution (Antonanzas et al., 2016; Zhang et al., 2015). Furthermore, they do not differentiate between two sets of data with equal variance and mean but different kurtosis and skewness. These considerations are especially important for system operators, as kurtosis is an indicator of the frequency of extreme events (Antonanzas et al., 2016). As a result, we also include standard deviation, kurtosis, and skewness in addition to RMSE and MAE. Finally, we do not include nightly observations in our evaluation as doing so would lead to misleading performance improvements.

The proposed metrics will be evaluated for all benchmark models for the one-hour, two-hour, three-hour, and four-ahead lead time, and as an average across lead times. The exception here occurs for ERA5 due to

having no specific lead time, as explained in Section 2.4. We also separate model forecast errors into different regimes, such as the hour of the day and the month of the year, to investigate the influence of seasonality.

In Yang et al. (2020), the authors propose a distribution-oriented approach with summary metrics based on a bias-variance decomposition and the Murphy-Winkler factorization (Murphy and Winkler, 1987). While this is currently the preferred approach for evaluating deterministic irradiance forecasts (Yang et al., 2020), this approach is impractical for the SARAH-2.1 dataset with resolution 512×512, yielding 262,144 irradiance forecasts at each time step. As a result, we follow this framework for our BSRN evaluation, where we use kernel conditional density estimation (KCDE) to estimate the conditional expectations from our deterministic forecasts. We kindly refer to Yang et al. (2020) for more details on the bias-variance decomposition, the Murphy-Winkler factorization, and KCDE.

4.2.1. Cloud Mask

As stated in Section 1, we include a cloud mask prediction for all models that directly predict the effective cloud albedo. These are IrradianceNet, smart persistence, and the *TV-L*^{1} optical flow. We follow the approach by Urbich et al. (2019), where CAL $= 0.025$ marks the threshold used to separate cloudy and clear-sky pixels. They found that this specific threshold yielded the highest probability of detection and lowest false alarm ratio. This separation essentially allows us to quantify errors in cloudy versus cloud-free situations, which is of utmost importance given the impact clouds have on solar irradiance, as stated in Section 1. We report irradiance errors from cloudy versus cloud-free situations in general and for different lead times in Table 4.

4.3. Ablation study

We experiment with several different settings for our IrradianceNet model in an ablation study. The first includes converting our deterministic IrradianceNet into the probabilistic setting using the techniques described in Section 3. It is not obvious a priori if such conversion should yield superior results compared to the deterministic setting, but the extension can still yield potentially useful information for electricity market participants. As an alternative to the patch-based cropping technique described in Section 3, we investigate downsampling our input images from 512×512 to 128×128, keeping our model architecture intact. We denote this model as IrradianceNetDownsampled, and we include it as part of our ablation study. It reduces the training and inference time considerably by only computing one forward and backward pass per image instead of 16 like in the patch-based approach. We note that IrradianceNetDownsampled is still being evaluated relative to the full resolution ground truth via a nearest-neighbor lookup to make it comparable to the original IrradianceNet model.

Another parameter of importance is the number of input time steps. Using four previous time steps as input can be relatively costly memory- and speed-wise, since we need to process them sequentially and save gradients for all steps. As a result, we experiment using only two-time steps as input. Finally, we also try removing the additional meta-input channels discussed in Section 2.2, which should make our model unable to encode the spatial and temporal context from the meta-features when making predictions. The results of these ablation studies will be presented in Section 5.5.

5. Results

We start by presenting our results based on the SARAH-2.1 dataset in Section 5.1, followed by a seasonal comparison in Section 5.2. Section 5.3 presents our results when using the Baseline Surface Radiation Network dataset as ground truth with the framework proposed by Yang et al. (2020). This is followed by an evaluation of the cloud mask prediction scheme in Section 5.4. Finally, we present the results of our ablation study in Section 5.5. We also present an example prediction of IrradianceNet versus ground truth for the SARAH-2.1 dataset in Fig. 2.

5.1. Overall Results

In Table 1, it can be seen that IrradianceNet achieves the lowest MAE and RMSE of 72.65 and 110.50, respectively, which are the primary metrics of interest. *TV-L*^{1} obtains the second-lowest MAE of 85.59 and RMSE of 137.68 as expected, followed by the smart persistence model with MAE of 89.25 and RMSE of 138.44. ERA5 lags behind with a MAE of 104.79, potentially due to its coarse resolution, but it is still superior to the naive persistence with an MAE of 173.83.

Looking at Table 2, it becomes apparent that IrradianceNet outperforms at all four lead times, which is initial evidence that a data-driven approach can better approximate the actual development and movement of the effective cloud albedo over multiple time steps compared to the benchmark models. However, the difference does become smaller at later time steps, as the difference between *TV-L*^{1} and IrradianceNet is only 10.01% and 5.32% at the three- and four-hour lead time. This suggests IrradianceNet is primarily focused on optimizing short-term dynamics and needs additional input information to make more plausible longer-term predictions. We note the relatively small difference between smart persistence and *TV-L*^{1}, which confirms the strong performance of the smart persistence model similar to previous studies Kumler et al. (2018).

Before investigating seasonal performance, we also include a figure of the mean absolute error on a pixel level over the entire test set in Fig. 3. This figure allows us to investigate spatial patterns in the error term, indicating regions where we should focus on improving our model further. Interestingly, we can clearly see a land-sea mask effect as errors are generally higher over land, on average. We hypothesize this is due to atmospheric convection having a different and potentially more unpredictable impact on the effective cloud albedo over land than at sea. Additionally, we see some relatively high errors over the Alpine region, potentially related to effects such as orographic lifting. Finally, it is also clear that we observe several border effects along the horizontal and vertical axis due to the patch-based approach discussed in Section 3.2. This is a known limitation of using this approach as stated in Section 3.2 and something we plan to address in the future with more sophisticated interpolation techniques. However, in a practical setting, a simple way to overcome this issue would be to strategically divide the patches to have borders lying in regions where there is little to no installed PV capacity.

5.2. Seasonal Evaluation

Seasonality in solar irradiance forecasting arises due to several reasons, such as sunrise and sunset on a short-term basis and different weather patterns observed in summer versus winter on a more long-term basis. In Fig. 4a, we group test performance into the various hours during the day. We exclude the naive persistence model here due to its inferior performance. It can be seen that IrradianceNet outperforms the benchmark models at all hours of the day. Comparing *TV-L*^{1} to smart persistence, we see the difference in performance is relatively consistent for all hours. Finally, looking at ERA5, we see it being inferior to the satellite benchmark models, especially during morning and afternoon hours, where dynamic weather patterns might be more frequent due to sunrise and sunset. As stated previously, this could be explained by its coarse resolution compared to the satellite-based benchmark models.

We include a similar figure for the monthly modality in Fig. 4b, which largely aligns with the results from the hourly modalities. We see IrradianceNet being superior in all months, particularly during the summer months, where solar irradiance is also relatively higher. Interestingly, we note that *TV-L*^{1} is superior to smart primarily, if not exclusively, during the summer months. This could be evidence that a

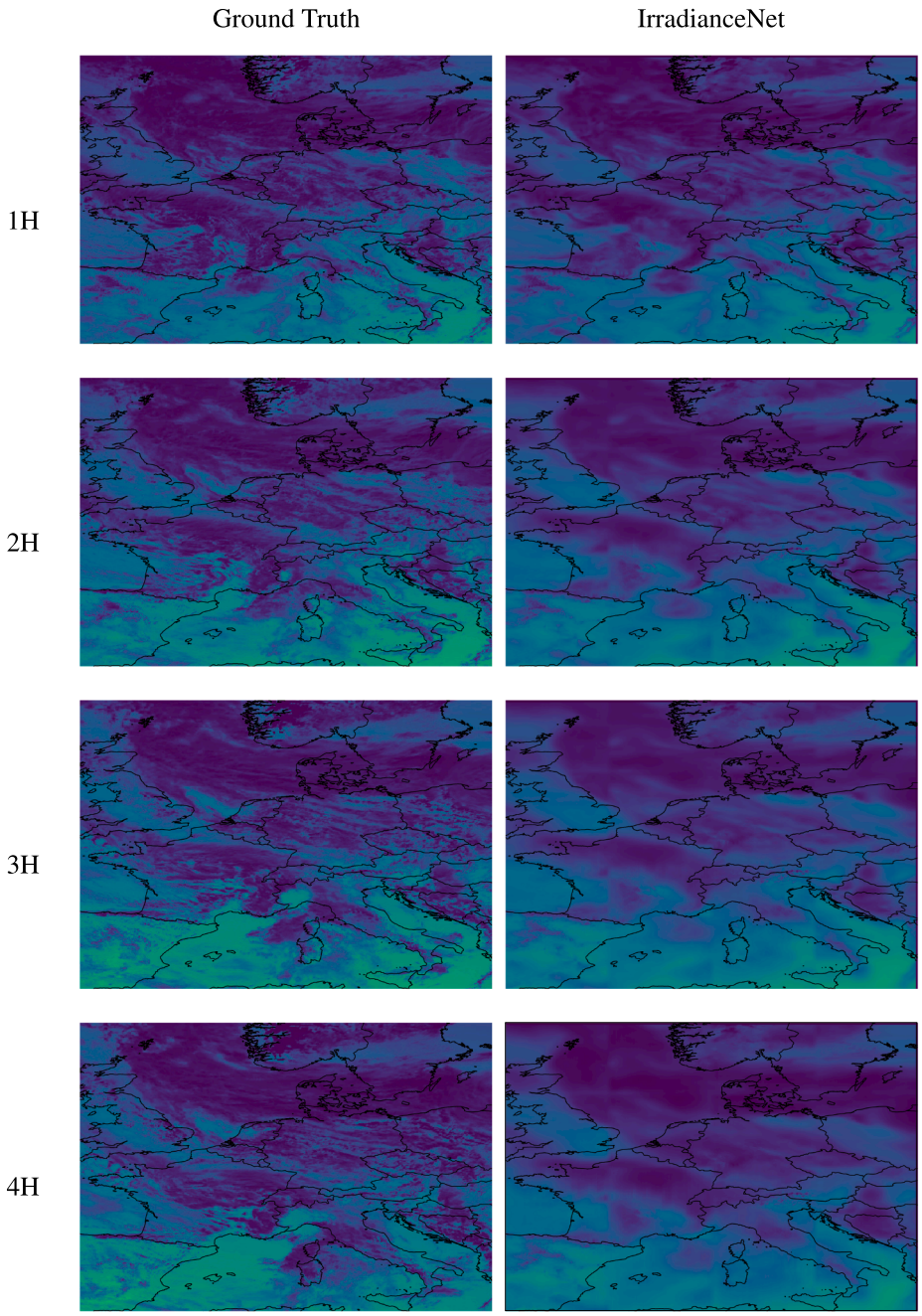


Fig. 2. Example forecast for IrradianceNet relative to the SARAH-2.1 ground truth. 1 H, 2 H, 3 H and 4 H refers to future forecast time relative to the initial start time at 2017–03-06 08:30.

Table 1
Performance evaluation with full-resolution models for 2016–2017 using the SARAH-2.1 dataset averaged across lead times. S.D. is the standard deviation of the forecasted irradiance, rRMSE (5-95th refers to root mean squared error normalized by the 5-95th percentile range.

Metric	Naive	Smart	TV-L^1	ERA5	IrradianceNet
MAE	173.83	89.25	85.59	104.79	**72.65**
RMSE	219.69	138.44	137.68	145.03	**110.50**
rRMSE ($5-95^{th}$)	0.27	0.17	0.17	0.19	**0.13**
S.D.	253.64	266.12	265.03	**228.22**	250.76
Skew	**0.04**	0.14	0.69	−0.06	−0.17
Kurtosis	**2.73**	5.09	5.73	2.77	4.66

Table 2
Mean absolute error for different lead times for 2016–2017 using the SARAH-2.1 dataset. NWP models are not included as these do not differentiate based on lead time. +% *TV-L*^{1} refers to the percentage improvement compared to *TV-L*^{1} as the reference forecast.

Model	1-Hour	2-Hour	3-Hour	4-Hour
Naive	102.74	164.46	215.70	254.15
Smart	70.36	87.92	100.50	108.79
TV-L^1	64.89	83.73	97.79	107.65
IrradianceNet	**51.39**	**64.16**	**88.01**	**101.92**
+% *TV-L*^{1}	**20.80%**	**23.37%**	**10.01%**	**5.32%**

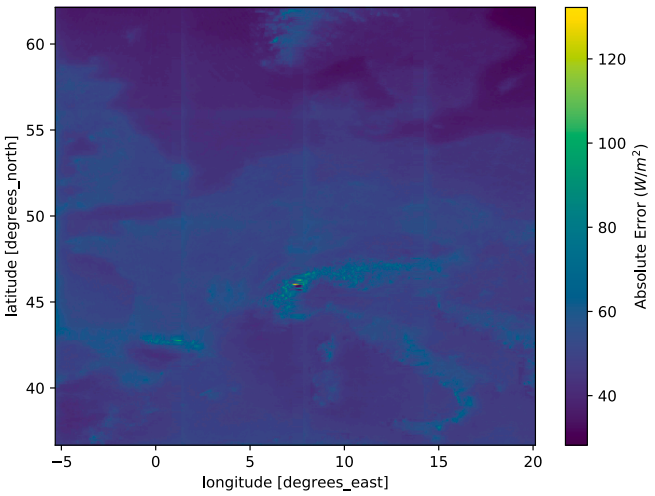


Fig. 3. Mean absolute error on a pixel level.

simple advection scheme such as *TV-L*[1] is mostly relevant during the summer months.

5.3. BSRN Evaluation

We now turn to the evaluation framework by Yang et al. (2020) for the four chosen BSRN sites. We have not included their conditional distribution visualization due to brevity. Otherwise, we follow their recommended framework. Fig. 5 shows the joint and marginal distributions of the four different BSRN stations. We notice that IrradianceNet tends to overforecast for all sites at low irradiance values (below 400) since the probability density contours lie above the diagonal. On the other hand, for mid-to-high irradiance values, IrradianceNet generally exhibits equal probability density above and below the density line, except for (b), where it continuously lies slightly above the diagonal at all irradiance values even though the standard deviation is smaller than (a), (c) and (d). These observations are perhaps not surprising given that the SARAH-2.1 dataset is not bias-corrected using ground observations, and we mentioned in Section 2 that a small and positive bias is present for these four sites.

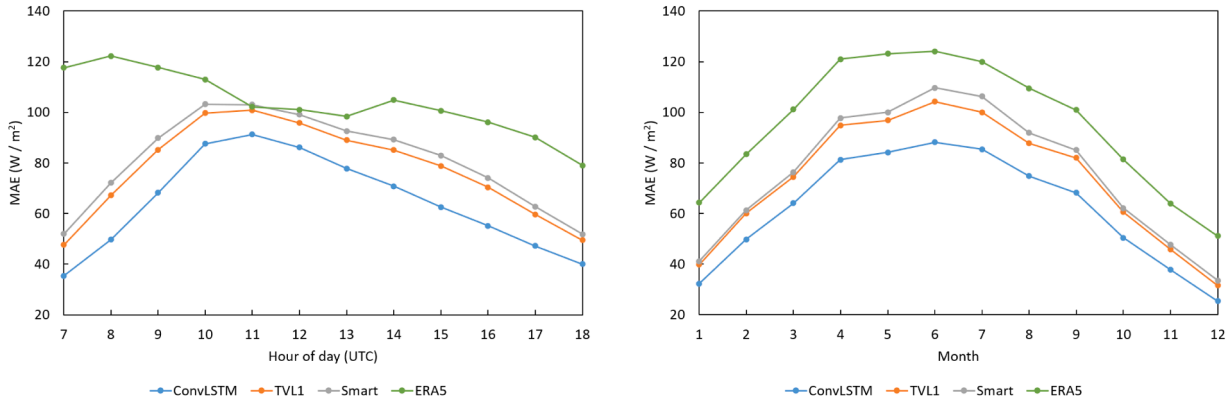
Next, we visualize overlaying the empirical cumulative distribution functions of forecasts versus observations at the four ground stations to investigate their marginal distributions more closely. This can be seen in Fig. 6. This figure largely confirms our observations from Fig. 5, as the ECDF of f lies below the ECDF of x for small irradiance values and generally drifts above for increasing irradiance values. Generally speaking, IrradianceNet seems to fit the ECDF of the observations x rather well, but we can make it more formal with the Kolmogorov–Smirnov (KS) two-sample test. The KS test statistic is formulated as the $D_n = \max|F_n(f) - F_n(x)|$, where $F_n(f)$ and $F_n(x)$ refer to the ECDFs of IrradianceNet and the ground observation, respectively, for all stations. At the 0.05 significance level, we reject the null of having equal distributions with p-values considerably below 0.01 for all sites.

As stated in Section 5.3, we complement the distribution-oriented figures with a table of summary measures based on bias-variance decomposition and the Murphy-Winkler factorization (Yang et al., 2020). We include this for IrradianceNet, the climatology-persistence model, *TV-L*[1], and ERA5. These results can be seen in Table 3.

IrradianceNet achieves the lowest RMSE and highest correlation coefficient for all sites, followed by *TV-L*[1], climatology-persistence, and ERA5. Looking into the different bias terms, we observe IrradianceNet having a relatively large bias in general, at least compared to climatology-persistence. This is an interesting observation, as it could reflect this bias in the SARAH-2.1 dataset that we mentioned in Section 2 and in Fig. 5, which is being propagated into the IrradianceNet model. Seeing this is also the case for *TV-L*[1] relative to ERA5 (except CNR), this is further evidence supporting this claim. If we dive into the conditional biases, we see IrradianceNet achieving relatively low type-1 bias for CAB and PAL compared to climatology-persistence, but a relatively high type-1 bias for CAR and CNR. For type-2 conditional bias, IrradianceNet achieves similar performance to *TV-L*[1] but considerably better than both climatology-persistence and ERA5. Finally, we also observe IrradianceNet being either the best or second-best for all sites for resolution and discrimination.

If we dive a bit deeper into differences among sites specifically for IrradianceNet, we see RMSE being lowest at CAR, where the correlation is also the highest with a value of 0.94. Correlation is roughly the same for the remaining three sites at around 0.91. We observe the highest unconditional bias at CAR, which aligns with our discussion for Fig. 5, where we saw IrradianceNet consistently overforecasting. For resolution and discrimination, we observe CAR and CNR achieving the best results.

Looking at RMSE for all benchmark models across sites, one is tempted to conclude that performance is highest at CAR and CAB. However, it is important to consider the relatively higher irradiance on average for some sites due to their geographical location. One customary solution to this problem is to normalize RMSE based on the mean of observations at each site (Yang, 2018). Doing this for IrradianceNet specifically, we obtain values of 0.273, 0.183, 0.223, and 0.269 for CAB, CAR, CNR, and PAL, respectively. As a result, we can conclude that IrradianceNet performs relatively better at CAR and CNR than CAB and



(a) Hourly Performance

(b) Monthly Performance

Fig. 4. Mean Absolute Error (MAE) for 2016–2017 for different temporal modalities. **(a)** shows the performance at different hourly modalities averaged across the time horizon and **(b)** shows the performance at different monthly modalities.

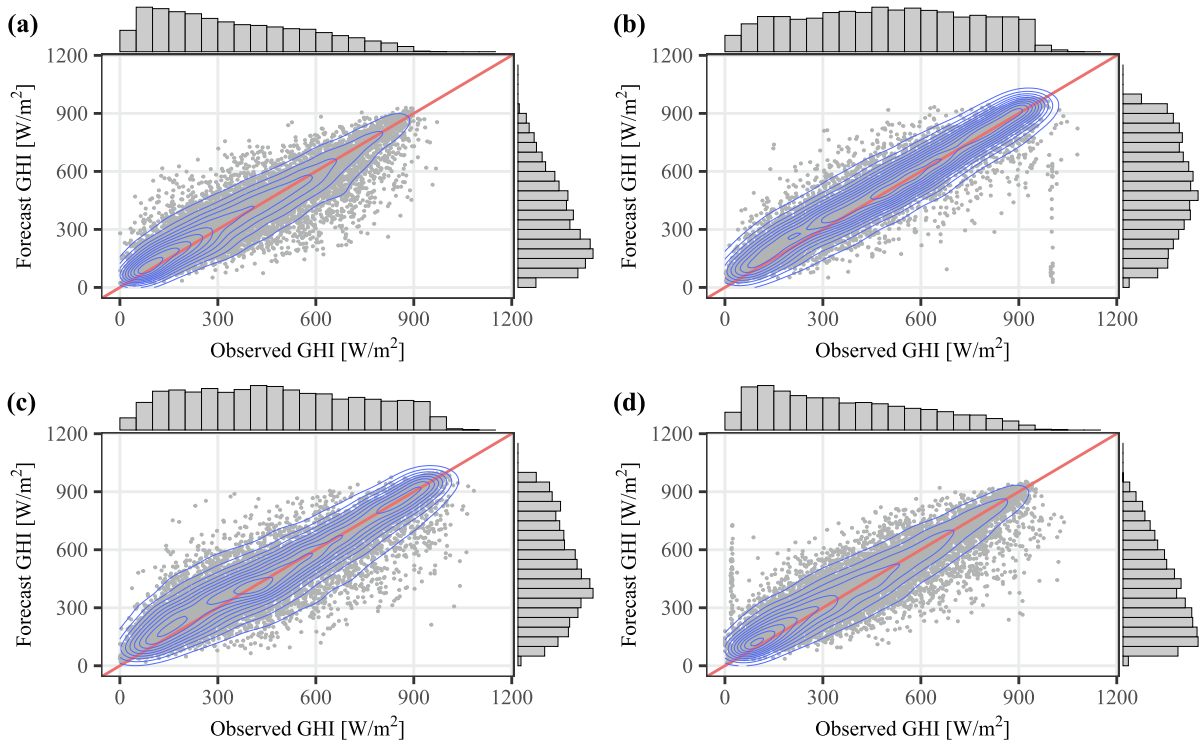


Fig. 5. Joint and marginal distributions of 1-h ahead hourly forecasts from IrradianceNet and benchmark models at CAB (51.9711°N, 4.9267°W), CAR (44.0830°N, 5.0590°W), CNR (42.8160°N, −1.6010°W) and PAL (48.7130°N, 2.2080°W). Contour lines deNote 2 D kernel densities.

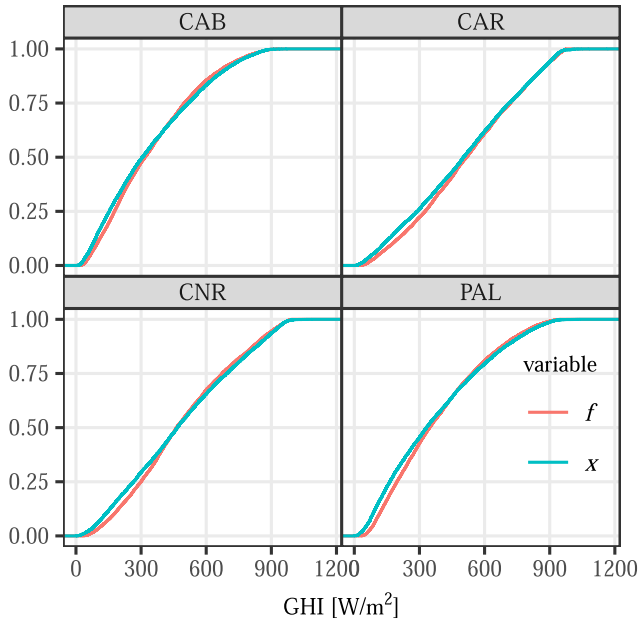


Fig. 6. Marginal distributions of observations and forecasts as described in Fig. 5.

PAL. From a meteorological point of view, this makes sense since CAB and PAL are characterized by more cloudy and unstable climates than CAR and CNR. Finally, we can conclude that IrradianceNet is consistently the best model among all benchmark models for the chosen BSRN sites.

5.4. Cloud Mask Evaluation

Next, we present the performance of our cloud mask prediction. It can be seen in Table 4 that cloudy pixels generally show higher errors compared to cloud-free pixels for all benchmark models, which was also demonstrated by Urbich et al. (2019). Comparing IrradianceNet to *TV-L*[1], we observe a considerable improvement in MAE for cloudy pixels of around 17.9 (18.1%) on average. On the other hand, cloud-free pixels show little to no difference between the three benchmark models on average. If we look more closely into the different lead times, we notice IrradianceNet being better than *TV-L*[1] for the one and two-hour lead time but inferior at the third and especially fourth lead time. This is an interesting result, as it suggests IrradianceNet might benefit from a more accurate clear-sky irradiance model at longer lead times. Nevertheless, we can conclude IrradianceNet is generally superior at forecasting and modeling the impact of clouds on surface solar irradiance, whereas cloud-free accuracy is largely unchanged on average compared to the benchmark models.

5.5. Ablation results

As stated in Section 4.3, the IrradianceNetDownsampled considerably improves inference time, as seen in Table 5 by a factor of 16 compared to the deterministic IrradianceNet-Det (Full) model. The reason is simply that we only need to process one 128 × 128 frame for the downsampled model instead of 16 times 128 × 128 frames for the full patch-based method even though they have same number of parameters. On the other hand, the performance is considerably worse, with an MAE of 108.27 for the downsampled model compared to 72.77 for the full model. For the probabilistic IrradianceNet-Prob, we observe more stable training than the deterministic IrradianceNet-Det, similar to other studies with the same probabilistic output encoding (Sønderby et al., 2020). Looking into Table 5, we see a considerable difference compared to the deterministic model. The IrradianceNet-Prob model achieves an MAE of 91.47 compared to 72.77 for the deterministic IrradianceNet-Det (Full) model. However, this does not invalidate using the probabilistic model, as probability estimates can still be extremely valuable to energy market participants, as discussed in Section 1. The

Table 3
Summary measures based on bias-variance decomposition and Murphy-Winkler factorization for the chosen benchmark models.

Model	Site	MSE	VAR(x)	VAR(f)	$\rho(f,x)$	Uncond. bias	Type-1 cond. bias	Resolution	Type-2 cond. bias	Discrimination
Irradiance-Net	CAB	94.75^2	225.57^2	210.78^2	0.91	4.86^2	11.34^2	204.89^2	36.46^2	191.71^2
	CAR	91.45^2	259.61^2	246.23^2	0.94	17.69^2	21.35^2	243.85^2	49.61^2	233.81^2
	CNR	107.47^2	260.47^2	240.80^2	0.91	6.11^2	10.99^2	237.43^2	45.87^2	220.26^2
	PAL	100.30^2	240.53^2	221.27^2	0.91	10.48^2	13.69^2	218.96^2	45.54^2	202.14^2
Climatology Persistence	CAB	121.43^2	225.57^2	184.50^2	0.84	3.31^2	11.51^2	190.32^2	71.49^2	156.12^2
	CAR	117.18^2	259.61^2	234.14^2	0.89	0.59^2	15.52^2	232.14^2	59.80^2	211.21^2
	CNR	130.45^2	260.47^2	217.78^2	0.87	1.93^2	15.75^2	225.97^2	76.88^2	190.52^2
	PAL	130.60^2	240.53^2	197.37^2	0.84	1.87^2	13.01^2	202.31^2	76.34^2	166.44^2
TV-L1	CAB	110.43^2	225.57^2	219.86^2	0.88	5.41^2	24.07^2	198.08^2	35.24^2	193.29^2
	CAR	105.61^2	259.61^2	252.34^2	0.92	17.50^2	29.04^2	238.89^2	49.80^2	234.42^2
	CNR	128.85^2	260.47^2	249.70^2	0.87	4.90^2	28.81^2	228.13^2	47.13^2	218.94^2
	PAL	117.62^2	240.53^2	229.35^2	0.88	9.70^2	23.50^2	211.04^2	45.61^2	201.82^2
ERA5	CAB	135.70^2	225.57^2	192.42^2	0.80	0.64^2	20.78^2	178.73^2	73.21^2	154.77^2
	CAR	142.13^2	259.61^2	220.33^2	0.83	4.48^2	21.77^2	215.66^2	83.41^2	187.88^2
	CNR	167.74^2	260.47^2	219.31^2	0.77	14.00^2	30.80^2	200.71^2	95.84^2	170.66^2
	PAL	149.65^2	240.53^2	208.29^2	0.78	0.64^2	28.85^2	187.02^2	78.99^2	164.74^2

Table 4
Performance evaluation in cloudy versus cloud-free regimes for 2016–2017 for models explicitly forecasting the effective cloud albedo at different lead times. Weights are based on number of observations at each lead time for the weighted averages.

	Mean Absolute Error	**Smart**	*TV-L*[1]	**Irradiance-Net**
Cloudy	1-Hour	82.83	74.87	57.79
	2-Hour	103.90	96.85	73.37
	3-Hour	113.79	112.95	99.20
	4-Hour	128.09	123.53	107.329
	Weighted Average	104.15	98.70	80.82
Cloud-Free	1-Hour	41.25	42.23	36.42
	2-Hour	53.05	53.46	43.61
	3-Hour	70.34	65.21	68.87
	4-Hour	68.55	74.30	90.14
	Weighted Average	56.23	56.57	55.93

Table 5
Ablation study using different configurations of our IrradianceNet model as discussed in Section 4.3.

Model	**MAE**	**RMSE**	**GPU Usage** (batch size = 1), MB	**Inference time (GPU)**, seconds
IrradianceNetDownsampled	108.27	153.86	728.77	0.0344
IrradianceNet-Prob	91.47	144.40	758.92	0.5808
IrradianceNet-Det (Only CAL)	80.44	122.61	727.01	0.5520
IrradianceNet-Det (t = 2)	75.95	118.06	508.75	0.4144
IrradianceNet-Det (Full)	72.77	110.50	728.77	0.5632

GPU usage and inference time are slightly worse for the probabilistic model due to the 240 categorical output encoding but still relatively fast. For IrradianceNet-Det (Only CAL), we disregard all the exogenous channels described in Section 2.2 and only use CAL as input. Here, we observe a decrease in performance of 7.67 in MAE compared to the full deterministic model. The GPU usage and inference time are mostly unchanged. Finally, using only two-time steps as input instead of four, called IrradianceNet-Det (t = 2), we observe a marginal decrease in performance of 3.18 MAE. Nevertheless, as the memory footprint is considerably lower, there could be valid reasons for considering this option as the performance is not considerably affected. In our analysis, we investigated how performance for this final ablation affected the various lead times. At the 1-h lead time, there was almost no change in performance with around 0.5% decrease compared to IrradianceNet-Det (Full). Arguably, this makes sense given the short lead time. At the 2-h lead time, the performance is decreased by around 1% and approximately 2% at the 3- and 4-h lead time.

6. Conclusions

This paper presented IrradianceNet, a satellite-based neural network for spatiotemporal forecasting of surface solar irradiance up to 4 h in the future over Europe. Our model is inspired by recent advances in spatiotemporal predictive models using deep learning, with novel modifications to the model architecture and input sources and encoding to specifically focus on modeling the effective cloud albedo, the primary uncertainty in current solar irradiance forecasts. We demonstrated superior performance relative to several benchmark models on data from 2016–2017, including the *TV-L*[1] algorithm and several persistence models recommended in the literature. This was validated on a) the satellite-based SARAH-2.1 dataset and b) four ground-based sites from the BSRN dataset evaluated using recommendations from extensive verification of deterministic forecasting studies.

For a) we observed a potential bias in the SARAH-2.1 dataset, which propagated into our data-driven model and led to overforecasting irradiance on average. We also demonstrated superior performance across different lead times, also when adjusting for hourly and monthly seasonality. As an additional experiment, we applied a simple cloud mask thresholding experiment. The findings implied that our IrradianceNet model being particularly good at modeling the impact of clouds on surface solar irradiance.

We hope IrradianceNet will lead to increased research into data-driven methods for satellite-based solar irradiance methods similar to the field of precipitation forecasting using deep learning. This can be extremely beneficial in many renewable energy applications involving solar power, particularly in regions where satellites might be the only data source available.

In future work, we would like to investigate the potential for solar power forecasting in an operational setting for multiple European countries. As stated in Section 2.1.1, using the irradiance dataset for operational forecasting is already a possibility. Also, conducting a comprehensive benchmark study to compare our probabilistic IrradianceNet with state-of-the-art probabilistic irradiance models could be

interesting, potentially with some more sophisticated output encoding techniques than the one presented in this paper. Finally, incorporating other data sources into our neural net, such as NWP data, could be interesting to investigate the potential for a longer forecasting time horizon or overcoming some of the land-sea and orographic effects we saw in Fig. 3.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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